Introduction To Programming

Python fundamentals

Universidad...

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Applied Physics | Course:

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Today...

Introduction to the course

1. Course info.
2. Brief Introduction to a Computer Machine.
3. Fundamentals.
4. What a program is and how an algorithm solves a problem.

The importance of two

Brief Introduction to Information Theory

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Brief Introduction to Information Theory

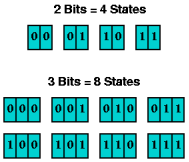
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The number 2 represents the fundamental minimum base for the existence and measurement of information.

Therefore, the number 2 is the absolute minimum required to build a complete arithmetic system.



Stibitz's "Model K" (1937): Relays & Batteries



Two distinct states: 0 and 1 bits

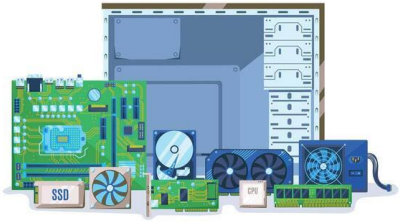
Computer Machine

Powering on the system

The Initial Spark:

When you press the start button of your computer, a signal sends voltage across your hardware components.

This sudden flow of electricity synchronizes the subsystem, preparing the physical architecture to communicate with the software.



Hardware Architecture

Inside the Brain

The CPU and the Kernel

Executing the First Instructions:

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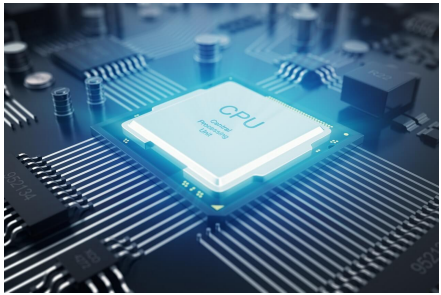
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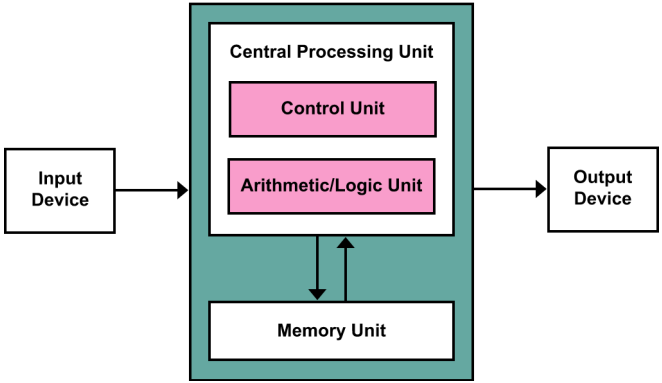
The CPU executes **millions of operations per second** just to initialize the graphic system and show you the login screen.



Modern Central Processing Unit (CPU)

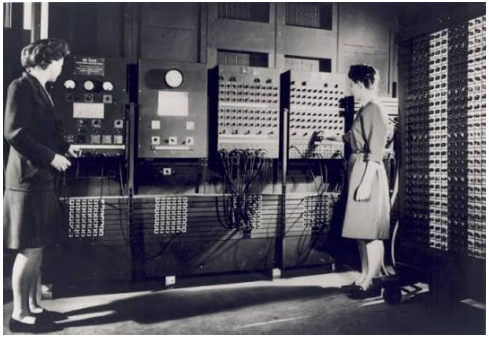
The thinks that computers do:

calculate billions of simple operations per second and remember thinks



Basic Machine Architecture

"The Eniac"



18,000 vacuum tubes, 30 tons.

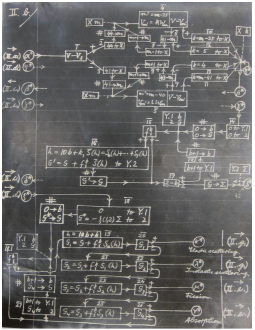


Diagram of subroutines, nested loops for the Eniac.

Nowadays...

Before: Low-Level (Assembly)

Coding was about micro-managing the circuits and memory.

```
MOV AX, 5 ; Initialize count
MOV BX, 3 ; Load modifier
ADD AX, BX ; Perform addition
PUSH AX ; Save result to RAM
```

→ Hard to write, easy to break.

Now: High-Level (Python)

Coding is about solving logical problems in human-centric terms.

```
# Fully abstracted
total = 5 + 3
print(total)
```

→ Focus on logic, not hardware.